FedHRA: A Joint Optimization Framework for Fast Convergent Decentralized Federated Learning in LEO Satellite Networks

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***Abstract*—Low Earth Orbit (LEO) satellites are playing an important role in earth observation, providing valuable images for training machine learning (ML) models used in tasks such as environmental monitoring and pattern recognition. However, due to unstable communication links and limited downlink bandwidth, it is economically impractical to transmit all raw images to ground stations (GSs) for model training. Federated learning (FL), a privacy-preserving distributed machine learning method, can reduce the communication overhead by exchanging model parameters. Generally, FL needs a fixed central server to aggregate the global model, which is challenging in LEO satellite networks, given the dynamic nature of satellite orbits. To overcome this, we propose a decentralized federated learning (DFL) framework that enables efficient model aggregation through satellite collaboration. Specifically, the proposed framework, named FedHRA, is based on model-agnostic meta-learning (MAML), which jointly optimizes hyperparameters and resource allocation to mitigate straggler effect and address statistical heterogeneity. Extensive numerical results on MNIST and CIFAR-10 datasets demonstrate that FedHRA achieves shorter learning time and higher model accuracy compared to the benchmark frameworks.**

***Index Terms*—Federated learning, low earth orbit satellites, statistical heterogeneity, model-agnostic meta-learning.**

I. Introduction

In RECENT years, an increasing number of Low Earth Orbit (LEO) satellites have been deployed, many of which are equipped with high-resolution imaging systems and advanced onboard sensors [1]. These satellites continuously capture vast amounts of Earth observation data, which are transmitted to ground stations (GSs) for further processing and machine learning (ML) model training to support critical applications, including earthquake early warning, environmental monitoring, and disaster prediction. However, due to the inherently limited bandwidth of satellite-to-ground communication links, offloading the complete set of raw data to GSs is often impractical [2]. Federated learning (FL) has recently emerged as a promising paradigm to alleviate these communication bottlenecks by training a ML model in each satellite using its local dataset and periodically uploading only model updates rather than raw data to a central server to construct a global model [3], [4], [5], [6]. Therefore, the communication overhead can be significantly reduced, and data privacy can be preserved, as sensitive sensing data remains onboard [7], [8].

Despite its advantages, the deployment of FL in satellite networks faces several practical challenges, including intermittent connectivity and limited onboard computational capabilities. Most existing studies adopt centralized FL architectures, where GSs act as parameter servers for model aggregation [9], [10], [11]. For example, the authors in [9] proposed an asynchronous FL framework for LEO satellite networks in which the GS performs centralized model aggregation. In [12], an asynchronous FL framework for collaborative training among satellite constellations was developed to address system heterogeneity. The work in [13] assigned model aggregation to satellites while offloading local training to ground devices. LEO satellites were further exploited as communication relays between ground stations in [14], where bandwidth optimization was used to improve communication efficiency under heterogeneous ground devices. A dynamic satellite selection mechanism for client aggregation was considered in [15], and a synchronous FL scheme within a satellite mega-constellation coordinated by an out-of-constellation parameter server was studied in [16]. Although employing a fixed server can offer benefits in terms of model accuracy and convergence stability, the high mobility and limited communication range of satellites make it infeasible to maintain persistent connections with a centralized server [17]. As a result, such centralized schemes are still hindered by long communication windows and significant propagation delays in satellite-GS links [18], which remains

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as a critical issue for data-intensive applications [19], [20], [21].

Decentralized federated learning (DFL) schemes have attracted increasing attention as a means to further decentralize the learning process across satellite constellations [22]. However, due to the relatively fixed orbital trajectories of LEO satellites, each satellite typically captures only a narrow category of Earth observation data, resulting in highly non-independent and identically distributed (non-IID) data across different orbits [23]. This statistical heterogeneity severely degrades the performance of the global model [24]. Although the parallel training architecture of DFL can improve the system efficiency, its performance is further constrained by heterogeneous onboard hardware and fluctuating inter-satellite link quality [25]. To mitigate these limitations, recent studies have explored optimizing satellite orbits and constellation sizes to reduce communication overhead [26]. In addition, collaborative computation strategies have been proposed to leverage distributed computing capabilities among satellites [27].

Nonetheless, existing studies on FL in LEO satellite networks have not sufficiently addressed the challenge of model generalization under non-IID data distributions. This limitation becomes critical in LEO constellations, where each satellite collects geographically and semantically distinct data due to orbit-specific sensing. To improve the generalization across such heterogeneous data, model-agnostic meta-learning (MAML) has been proposed as a meta-learning framework that enables models to rapidly adapt to new tasks by leveraging prior knowledge across distributed datasets [28]. By learning a well-initialized model, MAML not only accelerates convergence [29], [30], but also alleviates the impact of data heterogeneity and the straggler effect in decentralized training.

Given that both MAML and FL employ local gradient-based updates followed by global aggregation, they are naturally compatible [31]. This has led to the development of MAML-integrated FL frameworks, among which personalized federated learning (PFL) is a representative example. In particular, personalized federated averaging (Per-FedAvg) [32] enables each client to learn a personalized model while benefiting from global knowledge sharing. Although prior studies have applied Per-FedAvg in IoT scenarios [33], [34], they primarily focused on functional deployment without fully exploiting its advantages in reducing training latency or addressing resource constraints. In contrast, for resource-constrained LEO satellite networks, how to effectively integrate MAML-based personalization into decentralized FL remains largely unexplored.

Motivated by the above discussions, we propose a MAML-assisted DFL framework, termed FedHRA, which is tailored for satellite constellations with the goal of enhancing model generalization while optimizing training efficiency and resource utilization. The proposed FedHRA jointly optimizes hyperparameters and resource allocation to accelerate model convergence in LEO satellite networks. Unlike existing approaches that overlook the impact of training configuration and hardware heterogeneity, FedHRA explicitly accounts for both the learning dynamics and the practical constraints of the satellite environment. Specifically, FedHRA addresses two critical challenges: (i) the selection of hyperparameters, which significantly affects training latency and convergence behavior, and (ii) the allocation of limited satellite resources such as computation and transmit power under synchronous communication constraints. In each communication round, FedHRA calculates optimal hyperparameters and network parameters, e.g., computation and transmission power, for all participating satellites. Based on these settings, each satellite performs local model training, followed by parallel model aggregation with neighboring satellites. This process iterates until the global model achieves the target accuracy. By coordinating hyperparameter tuning with resource-aware scheduling, FedHRA reduces training time while preserving the convergence advantages of MAML. The proposed framework thus provides an efficient and scalable solution for deploying personalized federated learning in LEO constellations.

While federated learning for LEO satellite networks has indeed been investigated in recent literature, FedHRA differs from existing frameworks in architectural design. In particular, FedHRA couples MAML with decentralized aggregation structure that leverages intra plane inter-satellite links (ISLs) and inter plane ISLs. This design allows satellites to adapt rapidly to heterogeneous data. In addition, FedHRA achieves joint optimization of hyperparameters and resource allocation. Existing work typically fixes or adjusts hyperparameters independently, without considering their impact on system latency and convergence efficiency in dynamic satellite networks. FedHRA establishes the relationship between hyperparameters and model accuracy through theoretical modeling, thereby significantly reducing training delay while guaranteeing model accuracy. For a clearer comparison with prior studies, we summarize the key differences in Table I, emphasizing the novelty and contributions of the proposed approach.

The main contributions of this paper are summarized as follows:

- A novel MAML-based DFL framework, named FedHRA, is proposed for LEO satellite networks. The FedHRA leverages satellite image data to generate personalized models and enables efficient model aggregation through both intra-plane and inter-plane inter-satellite links (ISLs).
- A learning time minimization problem is formulated for FedHRA, which explicitly incorporates the impact of hyperparameters and network configurations. Specifically, we jointly optimize the sample data size, transmit power, and computational power of the satellites to minimize the learning time, while satisfying constraints on model accuracy and energy consumption.
- The convergence bounds of FedHRA is theoretically analyzed, and a quantitative relationship between hyperparameters and model accuracy is established. Based on this analysis, the original problem is decomposed into three sub-problems, each of which is solved using a coordinate descent approach to determine the optimal sample size, transmit power, and computation power.
- Extensive experiments on the MNIST and CIFAR-10 datasets demonstrate that FedHRA outperforms baseline FL frameworks in terms of both model accuracy and training efficiency with less learning time.

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TABLE I
COMPARISON OF RELEVANT WORKS WITH THIS PAPER

Ref.	DFL	Resource allocation	On-orbit collaborative computing	Hyperparameter optimization	Personalization model
[9]	✗	✗	✗	✗	✗
[13]	✗	✗	✓	✗	✗
[14]	✗	✓	✗	✗	✗
[27]	✓	✓	✓	✗	✗
[28]	✗	✗	✗	✓	✓
Proposed	✓	✓	✓	✓	✓

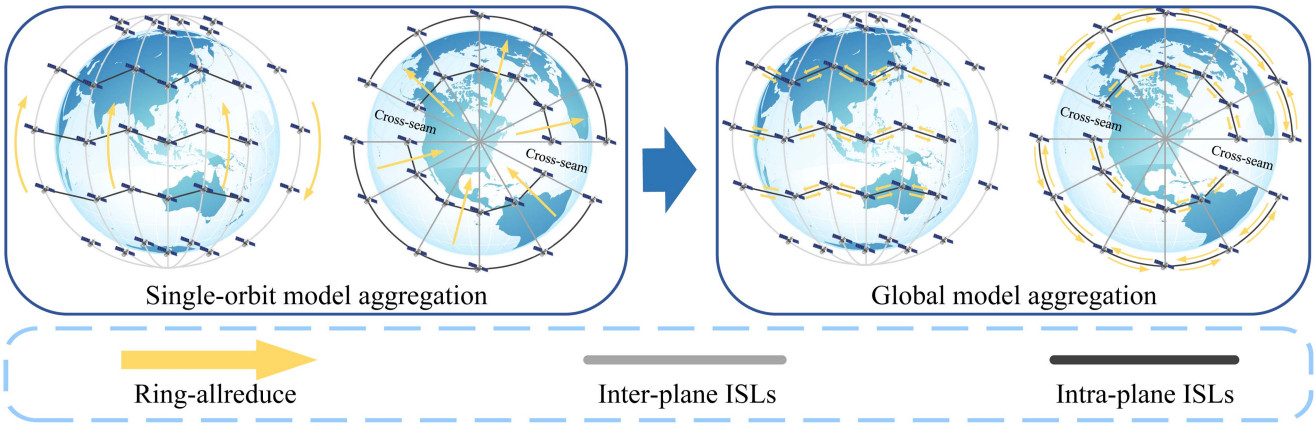


Fig. 1. Frontal view and top view of Walker Star constellation.

The rest of the paper is structured as follows. Section II introduces the system model. Section III focuses on problem formulation. Section IV analyzes the convergence and delay upper bounds, and obtains an approximate solution to the optimization problem by using coordinate descent method. Section V discusses the case with link failure. Section VI evaluates the performance of FedHRA, demonstrating its advantages and effectiveness. Finally, Section VII concludes this paper.

II. System Model

We consider a general Walker star LEO satellite network, where N satellites are evenly distributed across M equally spaced orbital planes, indexed by the set $\mathcal{M} = \{1, 2, \ldots, M\}$. The satellites within each orbital plane are denoted by $\mathcal{N}_m = \{1, 2, \ldots, N/M\}$. All orbital planes are deployed at the same altitude. Satellites establish both intra-plane ISLs within the same orbital plane and inter-plane ISLs between adjacent orbital planes [35]. As shown in Fig. 1, we assume cross-seam inter-plane ISLs[1] are infeasible due to high relative velocities and the Doppler effect caused by satellites in adjacent planes orbiting in opposite directions [36]. In mainstream LEO satellites constellations (e.g., Starlink, Telesat), satellites within the same orbital shell maintain constant velocities. The proposed FedHRA framework is composed of two parts, i.e., computing model and communication model, which are detailed below.

A. FedHRA Computing Model

In FedHRA, a satellite $n \in \mathcal{N}$ collects a local dataset $D_n = \{\boldsymbol{x} \in \mathcal{X}_n, y \in \mathcal{Y}_n\}$ with size $|D_n|$. The total dataset size across the LEO satellite network is given by $D = \sum_{n=1}^{N} D_n$. Each data sample consists of an input vector $\boldsymbol{x}$, such as image pixels, and a corresponding output scalar y, e.g., the label of the image. For the satellite n, the cross-entropy loss is defined as

$$l_n(\boldsymbol{w}) := \frac{1}{D_n} \sum_{(\boldsymbol{x},y)\in D_n} \sum_{j=1}^{J} Y(y=j) \log \widehat{Y}_j(\boldsymbol{x}, \boldsymbol{w}), \quad (1)$$

where J is the number of categories, Y is the sign function that takes the value 1 if the true label of the current sample is equal to category j, and 0 otherwise, $\widehat{Y}$ is the predicted probability that the current sample belongs to category j.

FedHRA leverages a limited number of data samples to learn an initial model via one or more gradient descent steps, enabling quick adaptation to new tasks. In this framework, each satellite is regarded as a distinct task, and it is assumed that each satellite updates its local model using its own loss function through a single gradient descent step. Accordingly, the personalized

[1]For the ISLs, communication is feasible if the line of sight is not obstructed by the Earth. Current research indicates that at altitudes higher than 80km above sea level, the quality of ISLs is not significantly degraded by atmospheric conditions [37]. Moreover, adaptive filtering and advanced modulation schemes have been used to improve the reliability of inter-plane ISLs [38]. Therefore, we assume that inter-plane ISLs other than the cross-seam inter-plane ISLs maintain stable connections.

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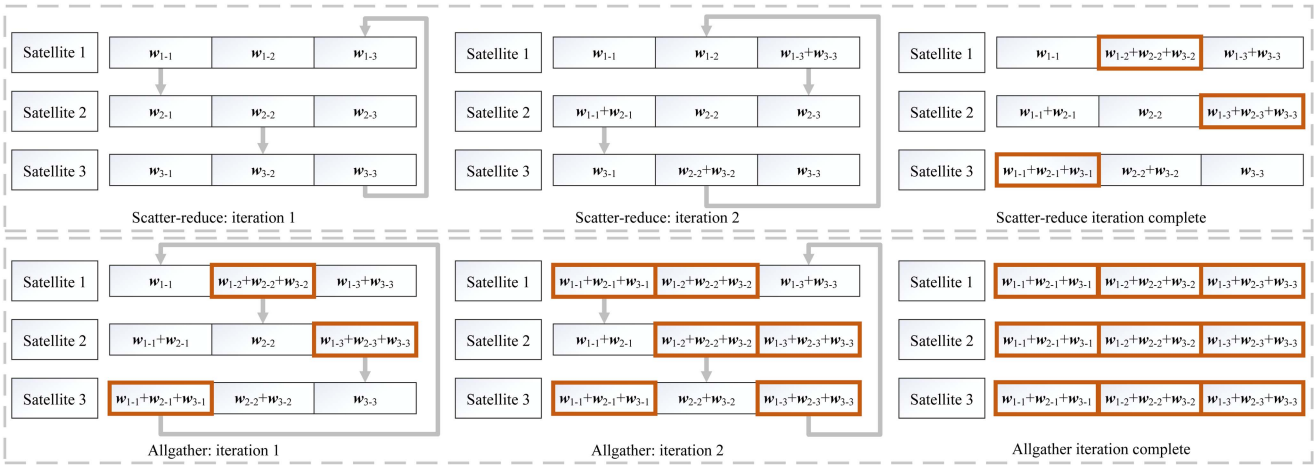


Fig. 2. Single-orbit model aggregation of LEO satellites, where N=9 and M=3.

model $L_n(\boldsymbol{w})$ for satellite n is defined as

$$L_n(\boldsymbol{w}) := l_n\left(\boldsymbol{w} - \alpha \nabla l_n(\boldsymbol{w})\right), \tag{2}$$

where α is learning rate. Unlike traditional FL, $L_n(\boldsymbol{w})$ is obtained after one-step gradient descent based on its own loss function for satellite n. The objective of FedHRA is to approximate a learning model trained on the union of all local datasets across satellites. To this end, each satellite performs local model updates independently and then engages in inter-satellite communication to exchange model parameters. Through iterative local updates and global aggregations, the DFL process aims to minimize the global loss function:

$$\text{(P1)}: \quad \min_{\boldsymbol{w}\in\mathbb{R}^m} L(\boldsymbol{w}) := \frac{1}{N}\sum_{n=1}^{N} L_n(\boldsymbol{w}). \tag{3}$$

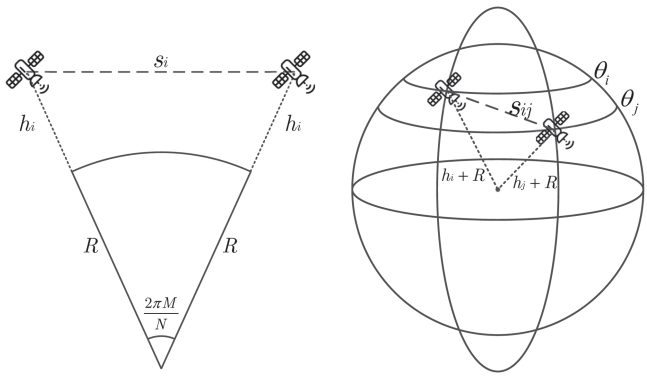
In the local update period, satellite n uses one-step stochastic gradient descent (SGD) on the personalized model to obtain the local model as follows [39]

$$\boldsymbol{w}_{k+1,n} := \boldsymbol{w}_k - \beta \nabla L_n(\boldsymbol{w}_k), \tag{4}$$

where $\boldsymbol{w}_k$ denotes the global model in round k, β is global model aggregation learning rate. The gradient descent method does not change the dimensions of the model parameters, so $\boldsymbol{w}$ trained by personalized models can be model-weighted averaged. At every round, computing the gradient $\nabla L_n(\boldsymbol{w}_k)$ by using all data points is often computationally expensive. Following [28], we instead compute an estimate gradient $\widetilde{\nabla} L_n(\boldsymbol{w}_k)$ as

$$\begin{aligned}\widetilde{\nabla} L_n(\boldsymbol{w}_k) = &\left(I - \alpha \widetilde{\nabla}^2 l_n(\boldsymbol{w}_k; D_n^h)\right) \\ &\times \widetilde{\nabla} l_n\left(\boldsymbol{w}_k - \alpha \widetilde{\nabla} l_n(\boldsymbol{w}_k; D_n^{in}); D_n^o\right),\end{aligned} \tag{5}$$

where D_n^{in}, D_n^o, and D_n^h are subsets of D_n for computing inner gradients, outer gradients, and hessian matrix, respectively. Note that $\widetilde{\nabla} L_n(\boldsymbol{w}_k)$ is a biased estimate of $\nabla L_n(\boldsymbol{w}_k)$, because $\widetilde{\nabla} l_n(\boldsymbol{w}_k - \alpha \widetilde{\nabla} l_n(\boldsymbol{w}_k; D_n^{in}); D_n^o)$ contains another stochastic gradient $\widetilde{\nabla} l_n(\boldsymbol{w}_k; D_n^{in})$. To reduce this bias, we assume that D_n^{in}, D_n^o, and D_n^h are mutually independent.



(a) two satellites in the same orbit (b) two satellites in different orbits

Fig. 3. Distance between two satellites.

In the LEO satellite network, the local model update time is computed for each satellite. In round k, the training delay of satellite n is defined as

$$t_{k,n}^u = \frac{\tau C_n d_{k,n}}{q_{k,n}}, \tag{6}$$

where τ refers to number of local iterations, C_n denotes the number of CPU cycles required to training 1-bit data for satellite n, $q_{k,n}$ is the CPU frequency of satellite n in round k, $d_{k,n} = D_{k,n}^{in} + D_{k,n}^o + D_{k,n}^h$ represents the total sampled data of satellite n in round k.

B. FedHRA Communication Model

After the local update, satellites in the same orbit exchange the model parameters through intra-plane ISL. The process operates as follow:

- *Step 1:* Scatter-Reduce Phase. As shown in Fig. 2, in this phase, each satellite n partitions its local model into N/M blocks and transmits the ith model block, denoted by $\boldsymbol{w}_{n-i}$, to its neighboring satellite. Upon receiving a model block, each satellite accumulates it with its own corresponding

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block and forwards the updated block to the next satellite in the ring.

- *Step 2:* All-Gather Phase. After $N/M-1$ rounds of scatter-reduce, each satellite holds a distinct fragment of the aggregated single-orbit model. In this step, satellites sequentially transmit their model fragments to the next satellite. Upon reception, each satellite replaces the corresponding local model block with the received one and forwards the updated model block to the next satellite. This procedure is depicted in Fig. 2.
- *Step 3:* Orbit-Level Model Reconstruction. After $N/M-1$ rounds of the all-gather operation, each satellite has received all model blocks and reconstructs the complete orbit-level aggregated model. The aggregated model for the mth orbital plane at communication round $k+1$ is expressed as

$$\boldsymbol{w}_{k+1}^{m}=\sum_{n=1}^{N}\sum_{i=1}^{N/M}\boldsymbol{w}_{n-i}, \tag{7}$$

- *Step 4:* Global Aggregation. Similar to the intra-orbit model aggregation process, the global model is obtained by aggregating the orbit-level models across all orbital planes. The global model at communication round $k+1$ is computed as

$$\boldsymbol{w}_{k+1}:=\frac{1}{\mathcal{M}}\sum_{m\in\mathcal{M}}\boldsymbol{w}_{k+1}^{m}, \tag{8}$$

where $\mathcal{M}$ denotes the set of all orbital planes, and $\boldsymbol{w}_{k+1}^{m}$ represents the aggregated model of the mth orbit.

When ISL cannot be established between two satellites in the cross-seam plane, the ground station can serve as a relay. Moreover, we assume that the ISLs maintain stable connections except for ISLs in the cross-seam plane. Therefore, the model $\boldsymbol{w}$ received by different satellites are the same. As shown in [40], the relationship between the computation power $p_{k,n}^{u}$ and the CPU frequency $q_{k,n}$ of satellite n in round k is expressed as

$$q_{k,n}=\sqrt{\frac{p_{k,n}^{u}}{\zeta_n}}, \tag{9}$$

where $\zeta_n\geq 0$ is the effective capacitance coefficient of satellite n, which is determined by the chip architecture. Since satellites maintain the same altitude in orbital i, the distance between two adjacent satellites in orbital i at an altitude of h_i is

$$\mathrm{s}_i=2\left(R+h_i\right)\sin\frac{\pi M}{N}, \tag{10}$$

where R denotes the radius of the earth, as shown in Fig. 3(a). Similarly, the angle between adjacent orbits are the same to ensure that the satellites cover the ground. Consequently, the distance between two satellites on adjacent orbits is obtained as [22]

$$\mathrm{s}_{ij}=\Bigg((h_i+R)^2+(h_j+R)^2-2(h_i+R)(h_j+R)\times\left(\cos\theta_i\cos\theta_j+\sin\theta_i\sin\theta_j\cos\frac{2\pi}{M}\right)\Bigg)^{\frac{1}{2}}, \tag{11}$$

where θ_i and θ_j denote the latitudes of two satellites, as illustrated in Fig. 3(b). To mitigate the Doppler effect, satellites in adjacent orbits are assumed to maintain a small latitude difference [41], and hence we set $\theta_i=\theta_j$. When the inter-satellite distance exceeds a certain threshold, the line-of-sight (LoS) link becomes obstructed. The maximum communication distance between two satellites is given by the LoS constraint [22], [41]:

$$s_m=\sqrt{h_i(h_i+2R)}+\sqrt{h_j(h_j+2R)}. \tag{12}$$

Based on (10) and (11), the transmit power of satellite n at round k is denoted by $p_{k,n}^{o}$. Accordingly, the signal-to-noise ratio (SNR) for intra-orbit and inter-orbit ISLs can be respectively expressed as

$$\mathrm{SNR}_{k,n}^{i}=\frac{p_{k,n}^{o}G_tG_r}{\overline{E_b}k_Brh_mQ}, \tag{13}$$

$$\mathrm{SNR}_{k,n}^{ij}=\frac{p_{k,n}^{o}G_tG_r}{\overline{E_b}k_Brh_{ij}Q}, \tag{14}$$

where G_t and G_r are the antenna gains of the transmitter and receiver, respectively. k_B is the Boltzmann constant, r is the total system noise temperature, and Q is the ISL margin. Moreover, $\overline{E_b}\triangleq\frac{E_b}{N_0}$ is the ratio of the received energy per bit to the noise density. Besides, $H_m=(\frac{4\pi s_i f}{c})^2$ indicates the free-space path loss between two satellites on the same orbit, and $H_{ij}=(\frac{4\pi s_{ij} f}{c})^2$ denotes the free-space path loss between two satellites on the adjacent orbit. Here c and f are the speed of light and carrier frequency, respectively.

Consequently, the transmission delay between two satellites on the same orbit is given by

$$t_{k,n}^{i}=\frac{Z\left(\boldsymbol{w}_k\right)}{B\log_2\left(1+\mathrm{SNR}_{k,n}^{i}\right)}, \tag{15}$$

and transmission delay between two satellites on the adjacent orbit is as follow:

$$t_{k,n}^{ij}=\frac{Z\left(\boldsymbol{w}_k\right)}{B\log_2\left(1+\mathrm{SNR}_{k,n}^{ij}\right)}, \tag{16}$$

where $Z(\boldsymbol{w}_k)$ represents the number of bits for $\boldsymbol{w}_k$, B is the transmission bandwidth. Based on (15) and (16), the transmission delay of satellite n in round k denotes as

$$t_{k,n}^{o}=t_{k,n}^{i}+t_{k,n}^{ij}. \tag{17}$$

We compute the energy consumption of each satellite. During the training process, the total energy consumption consists of model training energy e.g., computational energy, and model transmission energy e.g., communication energy. The energy consumption of satellite n in round k is denoted as $e_{k,n}$, which can be calculated as

$$e_{k,n}=\kappa t_{k,n}^{u}q_{k,n}^{3}+p_{k,n}^{o}t_{k,n}^{o}, \tag{18}$$

where κ is the effective switched capacitance determined by the chip architecture.

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III. Problem Formulation

Based on the preceding discussions, we model the system delay and formulate the FedHRA optimization problem. The total system delay in FedHRA consists of two main components: the local training delay and the communication (transmission) delay. For each communication round, the system delay is dominated by the slowest-performing satellite due to synchronous model aggregation. Therefore, the overall delay for completing K rounds of DFL can be expressed as

$$T = K \max_{n\in\mathcal{N}} \left\{t^u_{k,n}\right\} + K \max_{n\in\mathcal{N}} \left\{t^o_{k,n}\right\}, \tag{19}$$

where $t^u_{k,n}$ and $t^o_{k,n}$ denote the training and transmission delays of satellite n in round k, respectively, and K is the total number of DFL rounds.

The FedHRA optimization problem (P2) is formulated to minimize the system delay defined in (19), expressed as

$$(\text{P2}): \min_{d_{k,n},p^u_{k,n},p^o_{k,n}} T, \tag{20}$$

$$\text{s.t.} \ \frac{1}{K}\sum_{k=0}^{K-1} \mathbb{E}\left[\| \nabla L\left(\boldsymbol{w}_k\right)\|^2\right] \le \epsilon, \tag{20a}$$

$$0 \le e_{k,n} \le E^{\max}_n, \forall n \in \mathcal{N}, k = 0, \ldots, K-1, \tag{20b}$$

$$0 \le p^u_{k,n} \le P^{u\,\max}_n, \forall n \in \mathcal{N}, k = 0, \ldots, K-1, \tag{20c}$$

$$0 \le p^o_{k,n} \le P^{o\,\max}_n, \forall n \in \mathcal{N}, k = 0, \ldots, K-1, \tag{20d}$$

$$0 \le d_{k,n} \le D_n, \forall n \in \mathcal{N}, k = 0, \ldots, K-1, \tag{20e}$$

$$0 \le \mathrm{s}_i \le s_m, \tag{20f}$$

$$0 \le \mathrm{s}_{ij} \le s_m. \tag{20g}$$

Constraint (20a) enforces the convergence requirement of the learning process, ensuring that the expected squared norm of the gradient of the global loss function remains below the predefined threshold ϵ over K rounds. This condition aligns with the ϵ-approximate first-order stationary point (FOSP) criterion defined in Definition 1, which is a standard convergence metric for non-convex federated learning objectives. Constraint (20b) restricts the energy consumption of satellite n in round k to be within the maximum allowable energy budget $E^{\max}_n$. Constraint (20c) imposes an upper limit on the computational power of satellite n during round k, while Constraint (20d) bounds its transmit power. Constraint (20e) ensures that the selected local data size does not exceed the total amount of data available at satellite n. Finally, Constraints (20f) and (20g) represent line-of-sight (LoS) requirements, ensuring that intra-orbit and inter-orbit satellite communications occur within the maximum LoS distance.

IV. Solution to the Optimization Problem

Several fundamental challenges arise in the optimization of system delay in FedHRA, including: (i) the joint optimization over coupled satellite-specific variables $d_{k,n}, p^u_{k,n}, p^o_{k,n}$; (ii) the implicit dependency between these optimization variables and the resulting model accuracy; and (iii) the non-convex nature of the overall problem (20) with respect to the joint variable space. As a result, it is intractable to find a direct solution of problem (20). To address this challenge, we first conduct a theoretical analysis of convergence behavior. Building upon the theoretical foundation, we apply a coordinate descent-based decomposition strategy to decouple the original problem into three interdependent subproblems. Each subproblem focuses on optimizing a specific subset of variables, thereby facilitating an efficient computation of an approximate solution to the original non-convex formulation.

A. Convergence Analysis

To further analyze the convergence performance of the proposed optimization problem, we begin by establishing key properties of the loss function. Specifically, we adopt a set of standard assumptions commonly used in federated learning literature, consistent with those in Per-FedAvg [28]. Given that the local loss function $l_n(\boldsymbol{w})$ is a cross-entropy loss (as introduced in (1)), its gradient and Hessian are smooth. Accordingly, we make the following assumption.

Definition 1: A random vector $\boldsymbol{w}_\epsilon \in \mathbb{R}^m$ is called an ϵ-approximate FOSP if it satisfies $\mathbb{E}[\| \nabla L(\boldsymbol{w}_\epsilon)\|^2] \le \epsilon$.

Assumption 1: The gradient and Hessian matrix of $l_n(\boldsymbol{w})$ are Lipschitz continuous, satisfying

$$\| \nabla l_n(\boldsymbol{w}) - \nabla l_n(\boldsymbol{u}) \| \le \lambda \| \boldsymbol{w} - \boldsymbol{u} \|, \ \forall \boldsymbol{w}, \boldsymbol{u} \in \mathbb{R}^m, \tag{21}$$

and

$$\| \nabla^2 l_n(\boldsymbol{w}) - \nabla^2 l_n(\boldsymbol{u}) \| \le \rho \| \boldsymbol{w} - \boldsymbol{u} \|, \forall \boldsymbol{w}, \boldsymbol{u} \in \mathbb{R}^m, \tag{22}$$

where λ and ρ are Lipschitz constants.

Based on Assumption 1, the following lemma summarizes the smoothness and gradient estimation properties of the local and global loss functions.

Lemma 1: $L_n(\boldsymbol{w})$ satisfies the following properties, i.e.,

1) The average function $L(\boldsymbol{w}) = (1/N)\sum_{n=1}^N L_n(\boldsymbol{w})$ is λ_L-Lipschitz continuous with $\lambda_L := 4\lambda + \alpha\rho B$.

2) The bias and variance of the estimated gradient are bounded as

$$\left\| \mathbb{E}\left[\widetilde{\nabla} L_n(\boldsymbol{w}) - \nabla L_n(\boldsymbol{w})\right] \right\| \le \frac{2\alpha\lambda\eta_G}{\sqrt{D^{\text{in}}}}, \tag{23}$$

and

$$\mathbb{E}\left[\| \widetilde{\nabla} L_n(\boldsymbol{w}) - \nabla L_n(\boldsymbol{w})\|^2\right] \le \eta_L^2. \tag{24}$$

3) Variance is bounded as follow:

$$\frac{1}{N}\sum_{n=1}^{\mathrm{N}} \| \nabla L_n(\boldsymbol{w}) - \nabla L(\boldsymbol{w})\|^2 \le \gamma_L^2. \tag{25}$$

Lemma 1 depicts the smoothness of $L(\boldsymbol{w})$ and $L_n(\boldsymbol{w})$, quantifies the discrepancy between $\nabla L_n(\boldsymbol{w})$ and its estimated value $\widetilde{\nabla} L_n(\boldsymbol{w})$, and gives the deviation between $\nabla L_n(\boldsymbol{w})$ and $\nabla L(\boldsymbol{w})$.

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Supposing that Assumptions 1 and Lemma 1 are satisfied. The following theorem now provides the expected convergence results for the FedHRA described in Section III.

Theorem 1: If we have a sampled data size vector $\boldsymbol{d} \triangleq \{d_{k,n}\}_{K\times N}$, global number of rounds K, the optimal global loss $L(\boldsymbol{w}_\epsilon)$, and α within the interval $(0, 1/\lambda]$, the FOSP condition is given by

$$\frac{1}{K}\sum_{k=0}^{K-1}\mathbb{E}\left[\|\nabla L\left(\boldsymbol{w}_k\right)\|^2\right] \leq \frac{4\left(L\left(\boldsymbol{w}_0\right)-L\left(\boldsymbol{w}_\epsilon\right)\right)}{\beta K} + \left[\sum_{n=1}^{N}\left(\frac{D_n}{D}\right)^2\right]\left(2\beta\lambda_L\eta_L^2+\beta\lambda_L\gamma_L^2+\frac{\alpha^2\lambda^2\eta_G^2}{D^{\text{in}}}\right), \tag{26}$$

where $\boldsymbol{w}_k$ represents the average of $\boldsymbol{w}_{k,n}$.

The proof of Theorem 1 is provided in Appendix A. According to Theorem 1, we characterize the relationship between $\boldsymbol{d}$ and the predefined model accuracy ϵ. Combining Theorem 1 with constraint (20a) in problem (20), we aim to ensure that the right-hand side of (26) does not exceed ϵ. Therefore, we now give the following corollary.

Corollary 1: We assume that the conditions in Theorem 1 are satisfied. If we enable the global number of rounds $K = \mathcal{O}(1/\epsilon^3)$, the global learning rate $\beta = \mathcal{O}(\epsilon^2)$, and the number of data samples as $d_n = \mathcal{O}(1/\epsilon)$, then an ϵ-FOSP is obtained for problem (20).

Proof: $K = \mathcal{O}(1/\epsilon^3)$ and $\beta = \mathcal{O}(\epsilon^2)$ ensure that the order of magnitude of the first term on the right-hand side of (26) to be equal to $\mathcal{O}(\epsilon)$. Next, we analyze the second item on the right. Since D_n/D is independent of model accuracy ϵ, it is natural to consider the item $2\beta\lambda_L\eta_L^2 + \beta\lambda_L\gamma_L^2 + \alpha^2\lambda^2\eta_G^2/D^{\text{in}}$ for an ϵ-FOSP. By setting $D_n^{in} = D_n^h = D_n^o = \mathcal{O}(1/\epsilon)$ (i.e. $d_n = \mathcal{O}(1/\epsilon)$), $\alpha^2\lambda^2\eta_G^2/D^{\text{in}}$ dominates the value of $2\beta\lambda_L\eta_L^2 + \beta\lambda_L\gamma_L^2 + \alpha^2\lambda^2\eta_G^2/D^{\text{in}}$, because $2\beta\lambda_L\eta_L^2 + \beta\lambda_L\gamma_L^2 = \mathcal{O}(\epsilon^4)$ denotes the higher order infinitesimals of ϵ. Thereby, Corollary 1 is proved by combining the first and second terms.

Remark 1: The results in Corollary 1 provide a possible choice of hyperparameters. As long as the hyperparameters satisfy the conditions in Theorem 1 with respect to the magnitude of the model accuracy ϵ, other choices are also acceptable.

B. Transmission Delay Upper Bound Analysis

We analyze the upper bound of the overall transmission delay. By jointly examining the number of model aggregation rounds and the derived delay upper bound, we characterize how the system delay scales with the number of satellites. Specifically, the numbers of intra-orbit and inter-orbit aggregation rounds scale linearly with the ratio between the number of satellites N and the number of orbits M, which directly increases the communication overhead. Moreover, the derived upper-bound expression explicitly shows that the overall transmission delay grows as the number of satellites increases.

In the communication model, satellites exchange model parameters through ISLs. As indicated in the Steps 2 and 3 of FedHRA Communication Model, each satellite participates in $\bar{k}$ rounds of data exchange, given by

$$\bar{k} = \frac{N}{M} - 1. \tag{27}$$

Accordingly, the number of iterations required for model aggregation scales linearly with the total number of satellites N and inversely with the number of orbits M.

The upper bounds for the intra-orbit model transmission delay $t_{k,n}^i$ and inter-orbit model transmission delay $t_{k,n}^{ij}$ are

$$t_{k,n}^i \leq \frac{Z\left(\mathbf{w}_k\right)}{B\log_2\left(1+\frac{A_0N^2}{\pi^2M^2}\right)}, \tag{28}$$

and

$$t_{k,n}^{ij} \leq \frac{Z\left(\mathbf{w}_k\right)}{B\log_2\left(1+\frac{1}{2}A_0M\right)}, \tag{29}$$

where $A_0 = \frac{p_{k,n}^o G_t G_r}{\overline{E_b} k_B r Q}\left(\frac{c}{4\pi f}\right)^2 \frac{1}{4(R+h_m)^2}$, and h_m is the maximum altitude of the satellites.

Therefore, the upper bound of transmission delay $t_{k,n}$ is

$$t_{k,n} \leq \frac{Z\left(\mathbf{w}_k\right)}{B\log_2\left(1+\frac{A_0N^2}{\pi^2M^2}\right)} + \frac{Z\left(\mathbf{w}_k\right)}{B\log_2\left(1+\frac{1}{2}A_0M\right)}. \tag{30}$$

The proof of (30) is provided in Appendix B.

We further investigate the scaling behavior of the system delay with respect to the network size. As shown in (30), the total delay $t_{k,n}$ consists of two components: an intra-orbit term that scales with N^2/M^2, and an inter-orbit term that depends on M. Consequently, the system delay is jointly affected by the total number of satellites N and the number of orbits M. In particular, when N increases with a fixed M, the intra-orbit transmission delay dominates due to its quadratic dependence on N. In contrast, increasing M (e.g., deploying more orbits with fewer satellites per orbit) leads to growth of the intra-orbit delay, while reducing the inter-orbit delay.

C. Problem Decomposition

Following the results of Corollary 1, we reformulate constraint (20a) to clarify the relationship between the hyperparameters and model accuracy ϵ. From problem (20) and constraint (20b), we find that $\boldsymbol{p}^o$, $\boldsymbol{p}^u$, and $\boldsymbol{d}$ are coupled, but their impacts on both problem (20) and constraint (20b) are separable. Building upon this theoretical foundation, we apply a coordinate descent-based decomposition strategy to decouple the original problem into three interdependent subproblems. Each subproblem focuses on optimizing a specific subset of variables, thereby facilitating an efficient computation of an approximate solution to the original non-convex formulation. First, $\boldsymbol{d}$ is optimized while treating $\boldsymbol{p}^o$ and $\boldsymbol{p}^u$ as constants; next, $\boldsymbol{p}^o$ is optimized with $\boldsymbol{p}^u$ and $\boldsymbol{d}$ fixed; finally, $\boldsymbol{p}^u$ is optimized while keeping $\boldsymbol{p}^o$ and $\boldsymbol{d}$ constant. The iterative process repeats until convergence.

We provide theoretical guarantees for the proposed decomposition based optimization framework, including monotonic descent and block wise stationary, as detailed in Appendix C.

1) Sample Data Size Optimization: As a first step, we focus on the sample data size selection, which directly impacts

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both model convergence and system delay. The corresponding subproblem is formulated and solved as follows. Specifically, the sampled data size for satellite n is optimized in round k by solving problem (P3) defined in (31)

$$\text{(P3)}: \quad \min_{d_{k,n}} \max_{n} \left\{t^u_{k,n}\right\} + \max_{n} \left\{t^o_{k,n}\right\}, \tag{31}$$

$$\text{s.t. } d_{k,n} \geq \frac{1}{\epsilon}, \forall n \in \mathcal{N}, \tag{31a}$$

$$\kappa \frac{\tau C d_{k,n} \sqrt{p^u_{k,n}}}{\sqrt{\zeta_n}} + p^o_{k,n} t^o_{k,n} \leq E^{\max}_n, \forall n \in \mathcal{N}, \tag{31b}$$

$$0 \leq d_{k,n} \leq D_n, \forall n \in \mathcal{N}. \tag{31c}$$

Constraint (31a) is derived from Corollary 1, indicating that $d_n = \mathcal{O}(1/\epsilon)$ ensure an ϵ-FOSP. The substitution of $d_n = \mathcal{O}(1/\epsilon)$ by $d_{k,n} \geq \frac{1}{\epsilon}$ in constraint (31a) is reasonable because the accuracy of FedHRA is directly proportional to the sample data size, resulting in a smaller ϵ. Constraint (31a) is a contraction of (20a), further reducing the feasible domain of $d_{k,n}$. Thus, if the optimal $d^*_{k,n}$ exists in problem (P3), it is also feasible in problem (P2) and can be considered an optimal approximation to (20). Constraint (31b) represents the energy consumption limitation for satellite n. (31c) constrains the sample data size, ensuring it does not exceed the total number of data points generated by the satellite n.

According to (31), the optimal solution of $d_{k,n}$ for satellite n is determined by the lower bound $(1/\epsilon)$. However, this lower bound depends on the upper bounds defined by constraints (31b) and (31c), which are D^p_n and D_n, respectively. Specifically, D^p_n is given as $D^p_n = \sqrt{\zeta_n}/\kappa\tau C\sqrt{p^u_{k,n}} \times (E^{\max}_n - p^o_{k,n} t^o_{k,n})$. Based on these bounds, we consider two cases: $(1/\epsilon) > D_n$ and $(1/\epsilon) \leq D_n$. Furthermore, for each case, we analyze $(1/\epsilon) \geq D^p_n$ and $(1/\epsilon) < D^p_n$.

If satellite n generates a dataset that is larger than the lower bound of its model accuracy, i.e., $(1/\epsilon) \leq D_n$. In this case, we need to consider the relationship between $(1/\epsilon)$ and D^p_n further. If $(1/\epsilon) < D^p_n$, it means that the satellite n has enough energy to achieve the lower bound $(1/\epsilon)$. Therefore, the optimal sample data size is $d_{k,n} = (1/\epsilon)$. Conversely, if $(1/\epsilon) \geq D^p_n$, it means that the satellite n does not have enough energy to reach the lower bound $(1/\epsilon)$. According to constraint (31a), the model accuracy is monotonically increasing with respect to $d_{k,n}$, and the optimal sample data size is $d_{k,n} = D^p_n$. If the satellite n generates a dataset that is less than the model accuracy threshold, i.e., $(1/\epsilon) > D_n$, we need to further discuss the relationship between $(1/\epsilon)$ and D^p_n. When $(1/\epsilon) < D^p_n$, the optimal value of $d_{k,n}$ is D_n. Otherwise, the optimal value of $d_{k,n}$ is equal to $\min\{D_n, D^p_n\}$. The process of calculating $d_{k,n}$ is denoted as the Sample Data Size Optimization Process (SDSOP).

Complexity analysis: Satellite n obtains $d_{k,n}$ samples from the local dataset D_n in round k. This operation has the complexity of $\mathcal{O}(1/\epsilon)$, while (31) needs to compute the upper bound of $d_{k,n}$ with complexity $\mathcal{O}(1)$. Therefore, the complexity of P3 with N satellites is $\mathcal{O}(N/\epsilon)$.

2) Transmit Power Optimization: After optimizing the sampled data size, the next step focuses on communication resource allocation. In particular, we determine the transmit power of satellite n in round k by solving problem (P4) as specified in (32).

$$\text{(P4)}: \quad \min_{p^o_{k,n}} (\max_{n} \left\{t^u_{k,n}\right\} + \max_{n} \left\{t^o_{k,n}\right\}), \tag{32}$$

$$\text{s.t. } 0 \leq p^o_{k,n} \leq P^{o\max}_n, \forall n \in \mathcal{N}, \tag{32a}$$

$$\kappa t^u_{k,n} q^3_{k,n} + p^o_{k,n} t^o_{k,n} \leq E^{\max}_n, \forall n \in \mathcal{N}. \tag{32b}$$

Constraints (32a) and (32b) represent the transmit power and energy budget constraints for satellite n during round k, respectively. To solve problem (32), we first examine the relationship between the objective function and the transmit power variable $p^o_{k,n}$. By analyzing the monotonicity, we observe that $p^o_{k,n}$ and the resulting SNR are positively correlated, while SNR is inversely proportional to the transmission delay $t^o_{k,n}$. This indicates that the delay decreases with increasing $p^o_{k,n}$, suggesting a monotonic trade-off.

To ensure the energy constraint (32b) is satisfied, we further examine the derivative of the corresponding energy-related expression with respect to $p^o_{k,n}$. As shown in (33), the derivative of the transmit energy term is always non-negative and approaches zero asymptotically, confirming its monotonicity. Therefore, the optimal value of $p^o_{k,n}$ is obtained by taking the maximum transmit power that still satisfies all constraints.

$$\begin{aligned}
&\frac{d}{dp^o_{k,a}} \frac{p^o_{k,a} \times Z(\boldsymbol{w})}{B\log_2\left(1+\mathrm{SNR}^i_{k,a}\right)} \\
&= \frac{Z(\boldsymbol{w})}{B\log_2\left(1+\mathrm{SNR}^i_{k,a}\right)} \\
&\quad - \frac{Z(\boldsymbol{w}) \times \mathrm{SNR}^i_{k,a}}{B\left[\log_2\left(1+\mathrm{SNR}^i_{k,a}\right)\right]^2\left(1+\mathrm{SNR}^i_{k,a}\right)} \\
&> \frac{Z(\boldsymbol{w})}{B\log_2\left(1+\mathrm{SNR}^i_{k,a}\right)} \left(1 - \frac{\mathrm{SNR}^i_{k,a}}{\frac{\mathrm{SNR}^i_{k,a}}{1+\mathrm{SNR}^i_{k,a}}\left(1+\mathrm{SNR}^i_{k,a}\right)}\right) \\
&= \frac{Z(\boldsymbol{w})}{B\log_2\left(1+\mathrm{SNR}^i_{k,a}\right)}(1-1) = 0.
\end{aligned} \tag{33}$$

The inequality in (33) is due to the fact that $\log_2(1+x) > x/(1+x)$ $(x > 0)$, which can be easily verified using elementary calculus. By analyzing the first-order derivative of the $\mathrm{SNR}^i_{k,n}$ term, we observe that it is always positive. The same reasoning applies to the $\mathrm{SNR}^{ij}_{k,n}$ term. Consequently, the left-hand side of constraint (32b) is monotonically increasing with respect to $p^o_{k,n}$. This implies that constraint (32b) imposes an additional upper bound on $p^o_{k,n}$. Denote by P^o_n the maximum value satisfying constraint (32b), then the optimal transmit power $p^o_{k,n}$ can be expressed as

$$p^o_{k,n} = \min\left\{P^{o\max}_n, P^o_n\right\}. \tag{34}$$

The above procedure for calculating $p^o_{k,n}$ is referred to as the Transmit Power Optimization Process (TPOP).

Complexity analysis: Computing the upper bound for $p^o_{k,n}$ in problem (32) has a constant complexity of $\mathcal{O}(1)$. Consequently,

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solving problem (32) for N satellites results in an overall complexity of $\mathcal{O}(N)$.

3) Computation Power Optimization: Subsequent to the transmit power optimization, the computation power allocation for satellite n in round k is addressed by solving the optimization problem (P5) as specified in (35). This step aims to achieve an optimal balance between computational efficiency and the system's resource constraints.

$$\text{(P5)}: \quad \min_{p^u_{k,n}} (\max_n \{t^u_{k,n}\} + \max_n \{t^o_{k,n}\}), \tag{35}$$

$$\text{s.t. } 0 \le p^u_{k,n} \le P^{u\max}_n, \forall n \in \mathcal{N}, \tag{35a}$$

$$\kappa \frac{\tau C d_{k,n} \sqrt{p^u_{k,n}}}{\sqrt{\zeta_n}} + p^o_{k,n} t^o_{k,n} \le E^{\max}_n, \forall n \in \mathcal{N}. \tag{35b}$$

Constraints (35a) and (35b) denote the computation power constraint and energy consumption constraint. For fixed $d_{k,n}$ and $p^o_{k,n}$, it observed that increasing $p^u_{k,n}$ reduces the training time in round k. Therefore, the optimal value of $p^u_{k,n}$ is given by the minimum of the upper bound constraints (35a) and (35b). We denote the upper bound constraint of (35b) as $P^d_n = \sqrt{\zeta_n}/\kappa\tau C d_{k,n}(E^{\max}_n - p^o_{k,n} t^o_{k,n})$. Thus, the optimal value of $p^u_{k,n}$ is

$$p^u_{k,n} = \min \left\{ P^{u\max}_n, P^d_n \right\}. \tag{36}$$

The process of calculating $p^u_{k,n}$ is denoted as the Computation Power Optimization Process(CPOP).

Complexity analysis: Similar to problem (32), problem (35) computes $p^u_{k,n}$ for N satellites with a complexity of $\mathcal{O}(N)$.

D. Joint Optimization

After deriving the optimization strategies for the sampled data size $\boldsymbol{d}$, the downlink transmit power $\boldsymbol{p}^u$, and the computation power $\boldsymbol{p}^o$, a joint optimization framework is presented to solve the original problem via subproblem decomposition, as outlined in Algorithm 1.

In each communication round k, the sampled data size $d_{k,n}$ for each satellite n is first updated by solving the sample data size optimization subproblem, referred to as SDSOP in Algorithm 1. This step utilizes the transmit powers from the previous round as inputs, i.e., $p^u_{k-1,n}$ and $p^o_{k-1,n}$. Subsequently, by fixing $d_{k,n}$ and $p^u_{k-1,n}$, the downlink transmit power $p^o_{k,n}$ is optimized via TPOP. Then, given the updated $d_{k,n}$ and $p^o_{k,n}$, the computation power $p^u_{k,n}$ is determined by solving the computation power optimization subproblem through CPOP. The proposed optimization algorithm sequentially updates each variable group while keeping the others fixed. In the subsequent round $k+1$, the newly obtained $p^o_{k,n}$ and $p^u_{k,n}$ are used to re-optimize $d_{k+1,n}$, followed by corresponding updates to $p^o_{k+1,n}$ and $p^u_{k+1,n}$.

This iterative process continues until either the global model converges or a predefined training accuracy is reached. By leveraging the decomposability of the original non-convex optimization problem, the proposed algorithm achieves efficient and tractable joint resource allocation across communication and computation dimensions.

Algorithm 1: FedHRA Joint Optimization Framework.

Input: α, β, $E^{\max}_n$, $P^{u\max}_n$, $P^{o\max}_n$;
$p^o_{-1,n} := P^{o\max}_n$;
$p^u_{-1,n} := P^{u\max}_n$;
Initialization: A coordinating satellite is designated to receive the initial model $\boldsymbol{w}_o$ and broadcast it to all satellites in the LEO system;
for $k = 1$ *to* K **do**
  for *each satellite* $n \in \mathcal{N}$ **do**
    $d_{k,n} := \text{SDSOP}\left(p^u_{k-1,n}, p^o_{k-1,n}\right)$;
    $p^o_{k,n} := \text{TPOP}\left(p^u_{k-1,n}, d_{k,n}\right)$;
    $p^u_{k,n} := \text{CPOP}\left(p^o_{k,n}, d_{k,n}\right)$;
    Compute $\phi_{k,n} := \boldsymbol{w}_k - \alpha \nabla l_n(\boldsymbol{w}_k)$;
    Compute $\boldsymbol{w}_{k+1,n} := \boldsymbol{w}_k - \beta \nabla_{\boldsymbol{w}_k} L_n(\phi_{k,n}) = \boldsymbol{w}_k - \beta(\mathrm{I} - \nabla^2_{\boldsymbol{w}_k} l_n(\boldsymbol{w}_k)) \nabla_{\phi_{k,n}} l_n(\phi_{k,n})$;
  end
  Single Orbit Model Aggregation: Single orbit model aggregation with intra-plane ISL
  for *each satellite* $n \in \mathcal{N}_m$ *in same orbit* **do**
    $\boldsymbol{w}^m_{k+1} = \sum_{n=1}^{N} \sum_{i=1}^{N/M} \boldsymbol{w}_{n-i}$;
  end
  Global Model Aggregation: Global model aggregation using inter-plane ISL
  for *each satellite* $n \in \mathcal{N}$ *in different orbit* **do**
    $\boldsymbol{w}_{k+1} := \frac{1}{\mathcal{M}} \sum_{m \in \mathcal{M}} \boldsymbol{w}^m_{k+1}$;
  end
end

Overall complexity analysis: The overall time complexity of FedHRA mainly comes from the iterative execution of the three optimization subproblems and the model update process. Specifically, the SDSOP has a complexity of $\mathcal{O}(1/\epsilon)$ per round on each satellite, while the TPOP and the CPOP each have a complexity of $\mathcal{O}(1)$. In addition, each round requires local gradient updates and distributed model aggregation, contributing an extra constant overhead of $\mathcal{O}(1)$. Since the algorithm adopts a coordinate descent method to alternately optimize these variables, the total complexity must be multiplied by the number of iterations A. Considering all these components, the overall complexity can be expressed as

$$\mathcal{O} = 2\mathcal{O}(KN) + \mathcal{O}\left(KN/\epsilon\right) + \mathcal{O}(AKN), \tag{37}$$

where A is iteration number of the coordinate descent method.

V. Special Case With Link Failure

The established ISLs can be treated as quasi-stable within short aggregation intervals. However, in practical LEO satellite networks, ISLs may fail due to hardware faults or other unforeseen causes. To address ISL failures, we design a dynamic satellite-relay-based link strategy. Specifically, we consider a multi-hop, flow-based transmission scheme, where satellites forward data serially along each flow, while different flows can

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be active concurrently without intersection, i.e., each satellite participates in at most one flow due to physical communication resource constraints. Consistent with [42], we introduce two performance metrics as follows.

- *Availability:* A hop is said to be available if the link between the two communicating devices is not blocked by the Earth. A route is available if all hops along the route are available.
- *Coverage Probability:* The conditional coverage probability is defined as the probability that the received SNR at a single hop exceeds a predefined threshold, conditioned on the hop distance. The routing coverage probability is the probability that the received SNR at every hop along the route exceeds the threshold.

Given that the distance is s^o, the conditional coverage probability is defined as

$$P = \mathbb{P}\left[\mathrm{SNR}^o \geq \varepsilon \middle| s^o\right], \tag{38}$$

where ε is the coverage threshold. We consider that the communication is noise-dominated, and the power of co-channel interference is negligible. This is reasonable because of the satellite's narrow beam transmission mode and sufficient bandwidth to achieve orthogonal frequency allocation. This assumption has already been validated in [42].

Following [43], satellites maintain continuous connectivity by updating their available link in real time. When an ISL becomes unavailable, data can still be forwarded through neighboring satellites. The set of satellites in each flow can be denoted as $B(p)$. The delay of $B(p)$ is determined by the accumulated delays of the involved satellites n', which can be denoted as

$$T^o_{B(p)} = \sum_{n' \in B(p)} t^o_{n'}, \tag{39}$$

where $t^o_{n'} = \frac{Z(\mathbf{w})}{B\log_2(1+\mathrm{SNR}^o_{n'})}$. For the data transmission delay, it is determined by the slowest transmission flows in LEO satellite networks, we can denote it as

$$T^o = \max_{i=1,2,\ldots,p} T^o_{B(i)}. \tag{40}$$

By dynamically selecting available relay satellites, FedHRA employs multi-hop flow-based transmission scheme (i.e., $B(p)$) to ensure that model updates can continue to propagate even when some ISLs fail. The maximum flow delay T^o in (40) reflects the maximum delay. In this way, FedHRA can effectively handle transmission interruptions caused by ISLs failures in realistic environments with unstable ISLs, thereby ensuring the continuity and reliability of the decentralized federated learning process. In the subsequent experiments, we will evaluate the impact of this scheme on learning time and model accuracy, and compare it with baseline methods.

VI. Performance Evaluation

In this section, we conduct extensive experiments to evaluate the performance of FedHRA. Firstly, we verify its effectiveness in reducing learning time. Secondly, we demonstrate its convergence speed superiority through training loss and model accuracy. Finally, we validate its advantages in different network settings and dataset settings.

TABLE II
Simulation Parameters

Parameters	Value
Channel bandwidth, B	20 MHZ
Channel number per ISL	1
CPU cycle for training 1-bits data, C	500
Transmit power constraint, $p^{o\max}$	10 W
Computation power constraint, $p^{u\max}$	10 W
Effective capacitance coefficient, ζ	$[10^{-18}, 10^{-17}]$
Effective switched capacitance, κ	10^{-27}
CPU frequency, q	10^9
coverage threshold, ε	0 dB

A. Experimental Setup

1) Network Settings: We consider 100 satellites evenly distributed in 10 orbits and equidistantly distributed in each orbit. The orbital planes are positioned at an altitude of 600 km above the earth at a 90° inclination. Satellite Tool Kit (STK) simulation software provides the connectivity information between satellites. Other important simulation parameters related to the satellite are given in Table II. The simulations are based on the datasets MNIST and CIFAR-10. The training models for each dataset are described below:

- MNIST includes 10 classes of handwritten digit categorization datasets. The training model used for MNIST is a 2-layer DNN model with a hidden layer of 100 units.
- CIFAR-10 is a dataset containing 10 classes of RGB color images depicting common objects. The training model used for CIFAR-10 is LeNet-5 [44], which consists of two convolutional layers and three fully connected layers.

We divide the datasets into IID and non-IID distributions. For the IID setting, we randomly split the original training dataset into 100 equal parts and distribute one part to each satellite. For the non-IID setting, we first sort the data by label, then assign to each satellite samples from only 5 randomly selected classes. Furthermore, the datasets are randomly divided into 75% for training and 25% for testing.

2) Evaluation Scheme: To evaluate the performance of the proposed scheme, the following benchmarks are compared.

- FedAvg: FedAvg is the first framework proposed for FL and remains the most widely used and general FL algorithm. The sampled data size is fixed at 10 for all satellites, and the power limit for each satellite is assigned randomly from $1W$ to $10W$. Hyperparameter optimization and resource allocation are not performed.
- FedAvg-opt: In this scheme, the update steps are the same as in FedAvg without personalized model is used. The hyperparameters and resource allocation settings are the same as in FedHRA.

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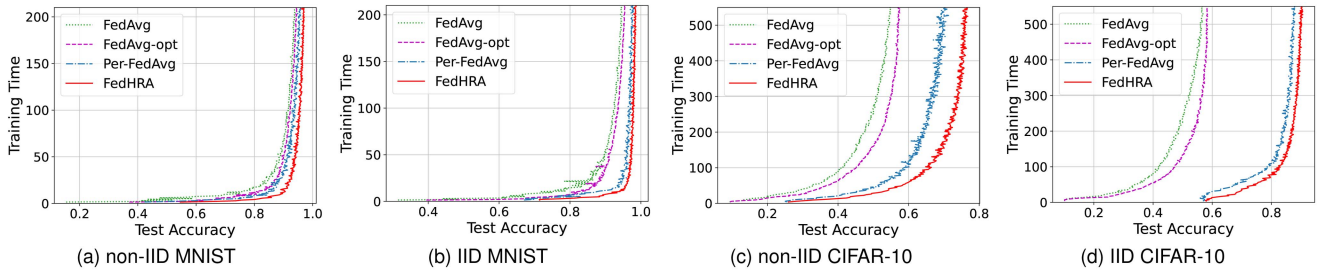


Fig. 4. Learning time comparisons using MNIST and CIFAR-10 data sets. As for non-IID Mnist and CIFAR-10 data sets, the number of labels is 5.

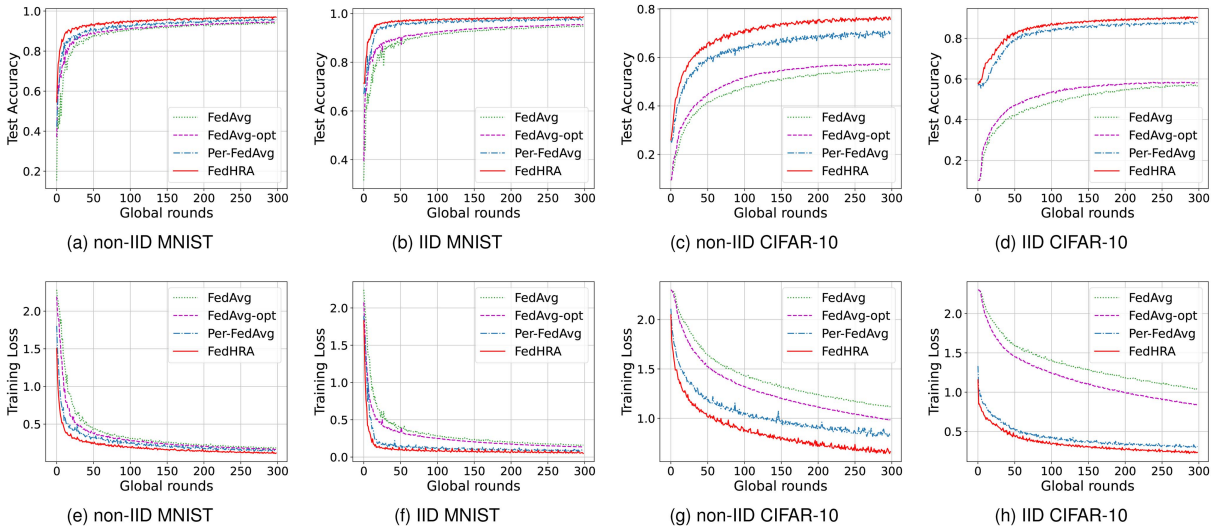


Fig. 5. Convergence performance using MNIST and CIFAR-10 data sets. As for non-IID Mnist and CIFAR-10 data sets, the number of labels is 5.

- Per-FedAvg: Per-FedAvg is the most widely used MAML-based FL framework. Neither hyperparameter optimization nor resource allocation is performed.
- FedAvg-ISL: In this scheme, we incorporate an ISL-failure mitigation strategy into the FedAvg framework to evaluate its performance.
- FedHRA-ISL: This scheme incorporates an ISL-failure mitigation strategy into the FedHRA framework.

B. Analysis of Convergence Efficiency and Learning Time

As illustrated in Figs. 4 and 5, the proposed FedHRA framework exhibits notable advantages in convergence efficiency across two datasets under both IID and non-IID settings. The following analysis elaborates on the performance from three key perspectives:

1) Accelerated Convergence With Competitive Accuracy: FedHRA consistently achieves the shortest learning time among all evaluated methods while maintaining competitive accuracy. This aligns well with the original design goal of FedHRA, that is, to minimize training time without sacrificing model performance. The experimental results corroborate our theoretical analysis and demonstrate the efficiency of the proposed approach in reducing convergence time.

2) Superiority of MAML-Based Approaches Under Non-IID Conditions: Both MAML-based methods (FedHRA and FedAvg-opt) outperform conventional FL frameworks (Per-FedAvg and FedAvg), with the performance gap being more pronounced under non-IID data distributions. This highlights the benefit of integrating meta-learning with federated optimization, and confirms that the joint optimization of hyperparameters and resource allocation plays a critical role in enhancing learning efficiency in heterogeneous environments.

3) Impact of Data Distribution on Learning Stability: Experiments further reveal that models trained on IID data exhibit more stable convergence and shorter learning times compared to those trained on non-IID data. This can be attributed to the improved alignment of local updates with the global objective, as each client's local model contributes more effectively to the aggregated model in the IID case, thereby accelerating the overall convergence process.

To further evaluate the convergence performance of FedHRA, we conduct a comprehensive comparison of its test accuracy and training loss against several baseline federated learning algorithms. Convergence performance is a critical metric in federated learning, as it reflects both the stability and efficiency of the learning process which is characterized by reduced training loss and improved test accuracy. As depicted in Fig. 5, FedHRA consistently outperforms baseline methods under both

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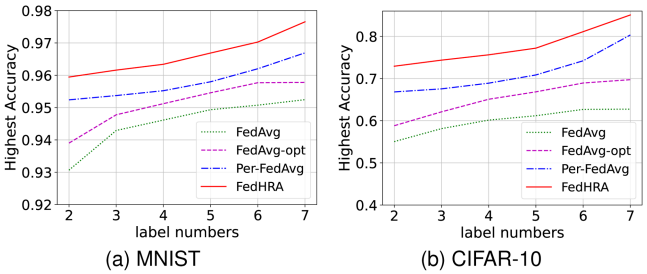


Fig. 6. Comparison of highest accuracy with different label numbers using MNIST and CIFAR-10 datasets.

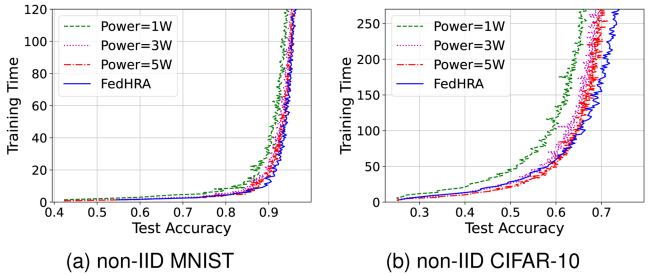


Fig. 7. Comparison of learning time under different power constraints using MNIST and CIFAR-10 datasets.

IID and non-IID data distributions. In the IID scenario, FedHRA achieves faster convergence and higher final accuracy than its counterparts. The advantages of FedHRA become more evident under the non-IID condition, where it maintains lower training loss and higher test accuracy throughout the training process. Particularly, on the CIFAR-10 dataset with CNN models, FedHRA demonstrates substantial improvements over baseline methods. It not only accelerates convergence but also yields superior final accuracy. These observations align well with the earlier results on learning time, further validating the effectiveness of FedHRA in enhancing convergence efficiency in federated learning environments.

C. Analysis of Accuracy Under Varying Non-IID Degrees

In this experiment, the number of labels assigned to each client is used to control the degree of data heterogeneity among LEO satellites. A smaller number of labels corresponds to a higher degree of statistical heterogeneity, thereby simulating a more challenging non-IID scenario. As shown in Fig. 6, we evaluate the accuracy performance of four schemes on the MNIST and CIFAR-10 datasets under different label configurations. The results reveal a consistent trend: as the number of labels increases, the model accuracy improves. This is because a higher label diversity enhances the informativeness of local datasets, which in turn improves the quality of model aggregation and global convergence in federated learning. Across all label settings, FedHRA consistently outperforms the other methods, demonstrating strong robustness against severe data heterogeneity. Even in highly non-IID settings, FedHRA maintains superior accuracy, highlighting its capability to adapt to diverse statistical distributions in LEO satellite communication systems.

D. Impact of Transmit Power on Learning Time Efficiency

The impact of transmit power on the learning efficiency of Per-FedAvg is illustrated in Fig. 7, where a clear trend is observed: as the transmit power increases from $1W$ to $5W$, the average learning time required to achieve the same accuracy level decreases progressively. This phenomenon is particularly evident in both the MNIST and CIFAR-10 datasets. The underlying reason for this improvement lies in the enhanced computational and communication efficiency afforded by higher power. With greater power allocated to the satellite, the system can process and transmit data more rapidly, thereby reducing latency and accelerating the convergence of the learning process. In contrast, FedHRA achieves high performance even without relying on high power.

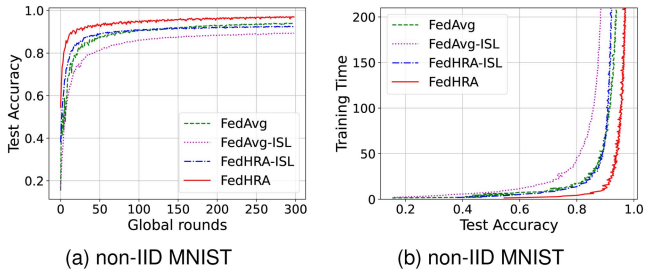


Fig. 8. Performance of model accuracy and training time with and without the ISL failures.

E. Impact of ISL Failures on Learning Time and Accuracy

We compare the scheme that includes the ISL failures handling strategy with the other frameworks. As shown in Fig. 8, the results indicate that incorporating strategies to address ISL failures affects system performance. The system latency of FedAvg-ISL and FedHRA-ISL is higher than FedAvg and FedHRA. When an ISL is interrupted, the system must rely on other satellites for relaying, which increases transmission latency and slows down the overall learning process. Therefore, FedAvg-ISL requires much more time to reach the same accuracy level. Moreover, FedAvg-ISL and FedHRA-ISL achieve visibly lower test accuracy than FedAvg and FedHRA because the latter rely on stable inter-satellite links for global aggregation. ISL failures disrupt model updates, leading to biased or incomplete aggregation.

VII. Conclusion

This paper proposed a fast convergent MAML-based DFL framework tailored for the LEO satellite networks, which is referred to as FedHRA. By incorporating personalized local models and leveraging intra-plane and inter-plane ISLs for collaborative model aggregation, FedHRA eliminates the reliance on a centralized server, enabling robust and scalable learning across distributed satellites. A theoretical convergence bound for FedHRA was derived to characterize the relationship between the algorithmic hyperparameters and the resulting model accuracy. Guided by this analysis, the original non-convex problem

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is decomposed into three subproblems, each corresponding to a key variable, and solved via an alternating optimization procedure to obtain an effective solution. Extensive experiments on MNIST and CIFAR-10 datasets under both IID and non-IID settings demonstrated the effectiveness of FedHRA in reducing convergence time while maintaining high model accuracy. In particular, FedHRA consistently outperforms two representative baseline methods, i.e., FedAvg and Per-FedAvg, highlighting the benefits of joint hyperparameter-resource optimization and personalized adaptation in non-stationary, heterogeneous environments. The proposed framework can facilitated deploying scalable and adaptive federated learning in satellite networks. Future work will explore the extension of FedHRA to LEO systems with focus on enhancing its robustness under adversarial conditions.

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